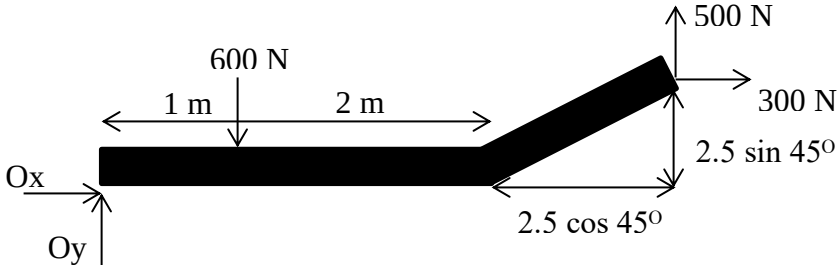
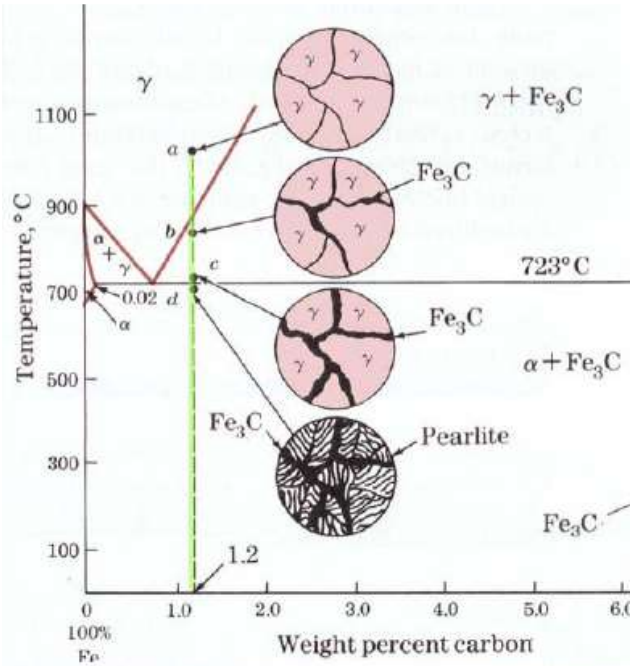


SET D

NO	ANSWERS				MARKS
1.(a)	- Metallic - Ceramics - Polymers				1 1 1 (total=3)
1.(b)		METAL 1 mark	CERAMICS 1 mark	POLYMERS 1 mark	1 mark for each box (total=12)
	DEFINITION	Metals are solid substances and have an excellent electric conductance. 1 mark	Produced by high temperature firing of inorganic, nonmetallic rocks and minerals. 1 mark	A long molecule consisting of a chain many repeating units. 1 mark	
	PROPERTIES	Metals have the property of ductility and malleability. 1 mark	Ceramic materials are brittle and hard. 1 mark	High strength to weight ratios, toughness, resilience, resistance to corrosion, lack of conductivity (heat and electric). 1 mark	
	EXAMPLE	Cast iron , rod iron, alluminium, etc. 1 mark	Clay, bricks, tiles, glass, cement 1 mark	Nylon, plastic,pvc 1 mark	

2.(a)	- Moment of a force about a point is equal to the sum of the moments of the forces' components about the point. OR - Moment of a force about a point or axis – a measure of the force to cause a body to rotate about the point or axis - $M = F \times d$ - Moment is defined as the product of the magnitude of force acting on the perpendicular distance from the force action line	2 2 1 (total=5)
2.(b)	FBD :  $M = Fd$ $\circlearrowleft \sum M_A = 600(1) + 300(2.5 \sin 45^\circ) - 500 (3 + 2 \cos 45^\circ)$ $= 600 + 530.33 - 2383.88$ $= 1253.55 \text{ N.m } (\circlearrowleft)$	Body : 1 Arrow : 1 Label : 1 Angle : 1 Distance : 1 (total=5) 1+1 1+1+1+1 1+1+2 2+2 (total=15)
3.(a)	<u>Traditional design:</u> - Sketches of drawing by using pencil, rulers, t-square and others basic tool. - Designed by a single person who invented a new product for customer. - Ingenuity of the design is well kept. - Non-reproducible. - More time consuming. - Not flexible. <u>Modern design:</u> - High generated complex designs by using design software such as CAD. - Designed by a team of people mostly to improve current product designs to satisfy customer needs. - Response to general and specific requirements. - Adaptable to changes.	1 1 1 1 1 1 1 1 1 1 1

3(b)	Identify the need Define the problem Search for Information Identify Constraints Specify Evaluation Criteria Generate Alternative Solutions Engineering Analysis Optimization Decision Design Specifications Communication	1 1 (total=12) (total=8)																				
4.(a)	<u>Center of gravity for system of particles</u> locates the <u>resultant weight</u> of a system of particles. <u>Center of mass for a system of particles</u>, provided acceleration due to gravity g for every particle <u>is constant</u>. <u>Center of mass for a rigid body</u> is composed of an infinite number of particles. <u>Centroid for a body</u> defines the <u>geometric center of object</u>.	2 3 2 2 (total=9)																				
4 (b)	Centroid: <table><tr><th>Segment</th><th>$A(\text{cm}^2)$</th><th>$\bar{y}(\text{cm})$</th><th>$\bar{y}A(\text{cm}^3)$</th></tr><tr><td>Triangle</td><td>$\frac{1}{2}(3)(3) = 4.5$</td><td>$\frac{2}{3}(3) = 2$</td><td>9</td></tr><tr><td>Square</td><td>$3 \times 3 = 9$</td><td>$\frac{1}{2}(3) = 1.5$</td><td>13.5</td></tr><tr><td>Circle</td><td>-0.785</td><td>1.5</td><td>-1.1775</td></tr><tr><td>Total</td><td>12.715</td><td></td><td>21.3225</td></tr></table> $\therefore \bar{Y} = \frac{\sum \bar{y}A}{\sum A} = \frac{21.3225}{12.715}$ $= 1.677\text{cm}$ 1M 1M	Segment	$A(\text{cm}^2)$	$\bar{y}(\text{cm})$	$\bar{y}A(\text{cm}^3)$	Triangle	$\frac{1}{2}(3)(3) = 4.5$	$\frac{2}{3}(3) = 2$	9	Square	$3 \times 3 = 9$	$\frac{1}{2}(3) = 1.5$	13.5	Circle	-0.785	1.5	-1.1775	Total	12.715		21.3225	2 2 2 2 1 1
Segment	$A(\text{cm}^2)$	$\bar{y}(\text{cm})$	$\bar{y}A(\text{cm}^3)$																			
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Circle	-0.785	1.5	-1.1775																			
Total	12.715		21.3225																			

4(c)	Fill the empty spaces : <table border="1" data-bbox="626 170 1276 506"> <thead> <tr> <th>Segment</th><th>Area, $\tilde{A}$ (mm²)</th><th>x (mm)</th><th>$\tilde{A}x$ (mm³)</th></tr> </thead> <tbody> <tr> <td>Rectangle</td><td>1200</td><td>10</td><td>12000</td></tr> <tr> <td>Triangle</td><td>540</td><td>30</td><td>16200</td></tr> <tr> <td>Total</td><td>1740</td><td>28200</td><td></td></tr> </tbody> </table>	Segment	Area, $\tilde{A}$ (mm ²)	x (mm)	$\tilde{A}x$ (mm ³)	Rectangle	1200	10	12000	Triangle	540	30	16200	Total	1740	28200			2 2 2
Segment	Area, $\tilde{A}$ (mm ²)	x (mm)	$\tilde{A}x$ (mm ³)																
Rectangle	1200	10	12000																
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Total	1740	28200																	
5(a)	- Hypoeutectoid steel having below the 0.77% carbon - Hypereutectoid steel having above the 0.77% carbon - A hypoeutectoid steel has a carbon concentration less than the eutectoid; on the other hand, a hypereutectoid steel has a carbon content greater than the eutectoid		2 2 2																
5(b)	- 		9																